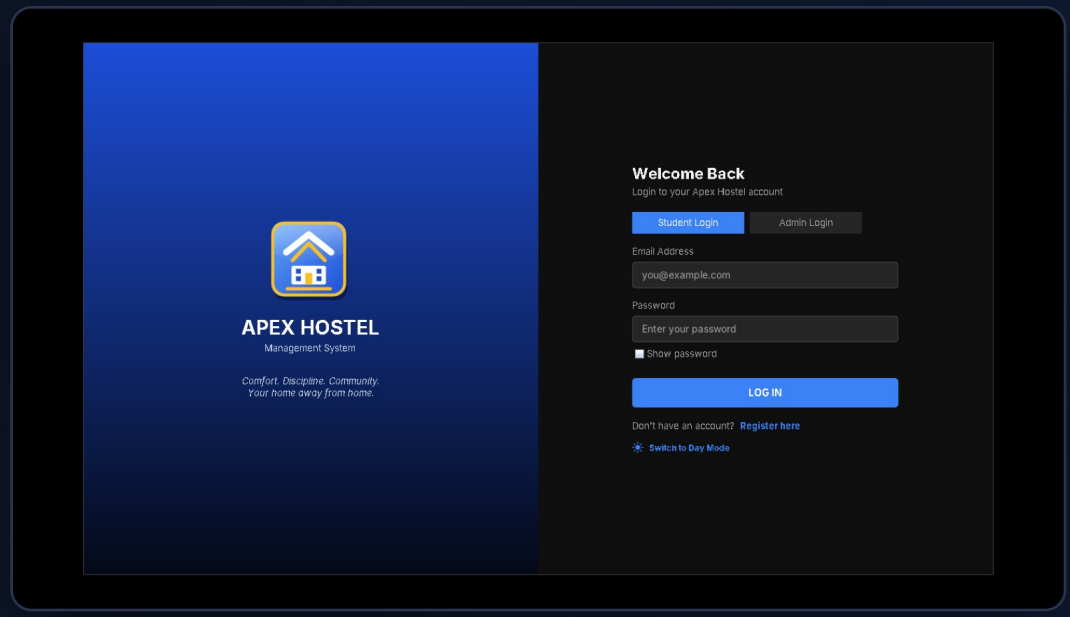
PROJECT PRESENTATION

Apex Hostel Management System

A Java Swing and MySQL desktop app where students view rooms and fees, and the admin runs the hostel.



OVERVIEW

One system, two roles



Student

Registers, logs in, views which rooms are free, checks fee challans, files complaints and reads notices.



Admin

Manages students, allocates rooms, generates fee challans, resolves complaints and posts notices.



MySQL

Keeps every record. Tables are created on the first run, with 14 sample rooms ready to use.

TECHNOLOGY

How it is built

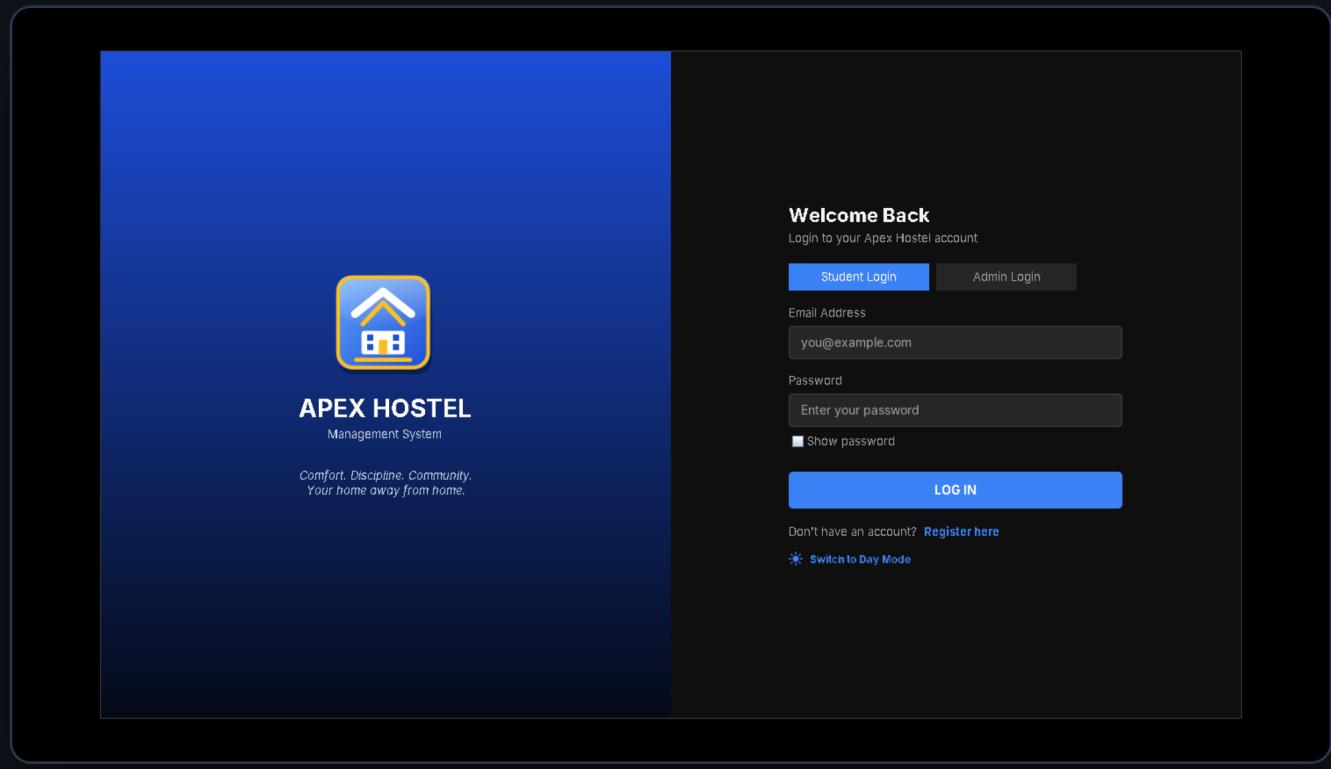
Layer	What we used
Interface	Java Swing with custom painted buttons, tables and icons
Database	MySQL through JDBC (Connector/J)
Security	Salted SHA-256 password hashing
Themes	Dark and Day mode, remembered between runs
Code	21 separate class files in one src folder

Start it with run.bat, or open ApexHostelSystem.java in IntelliJ and press Run.

ACCESS

Secure login and registration

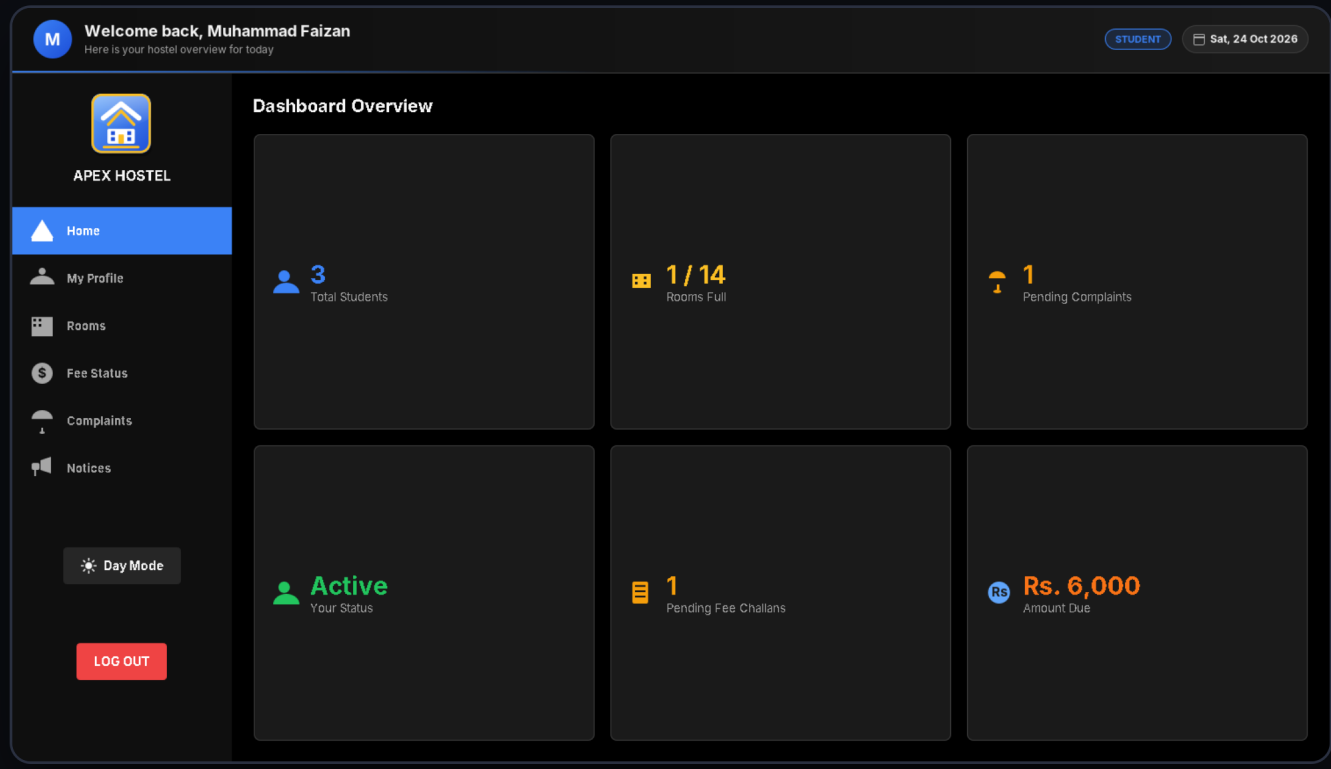
- Separate Student and Admin login tabs
- New students register with CNIC and department
- Passwords stored as salted SHA-256 hashes
- Old plain passwords upgrade at next login



STUDENT SIDE

Student dashboard

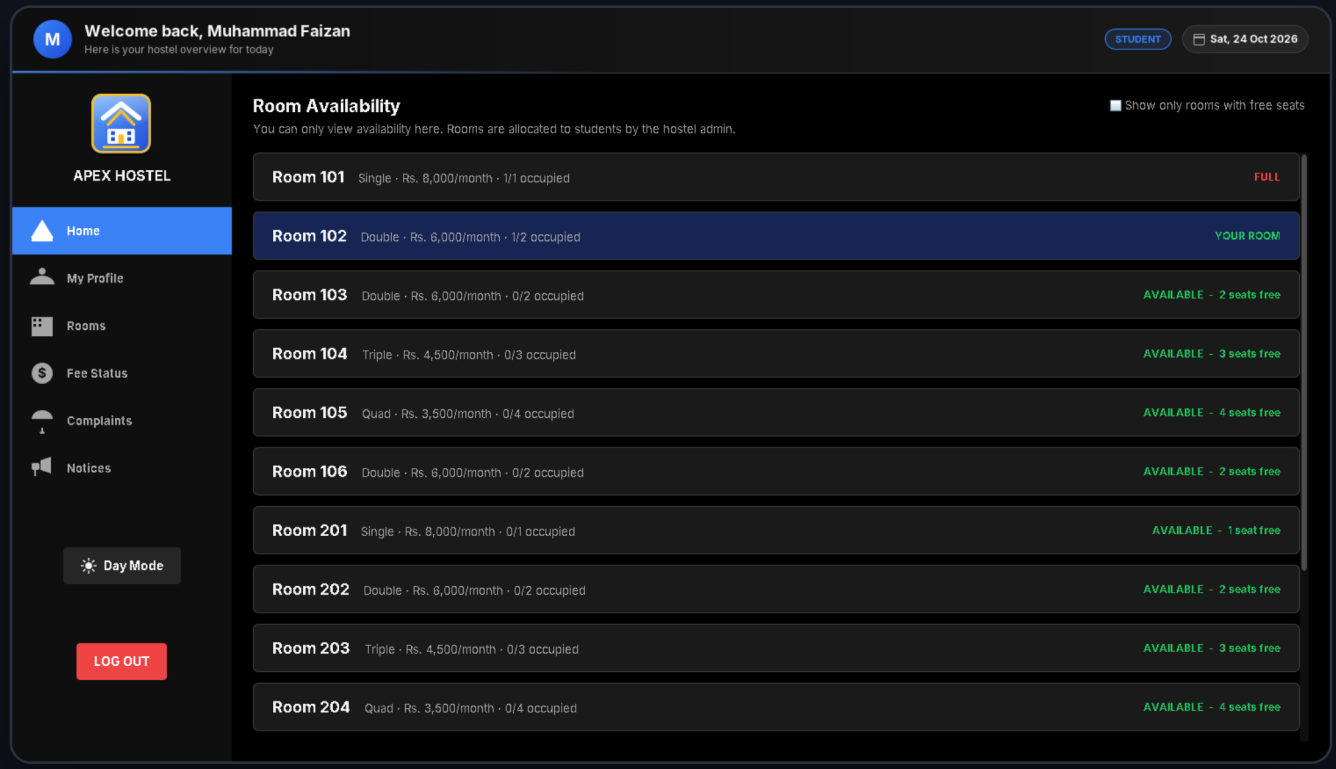
- Overview cards: rooms, complaints, fees due
- Profile page and password change
- Fee status with challans to view and pay
- Complaints and notices in one sidebar



STUDENT SIDE

Rooms: view only

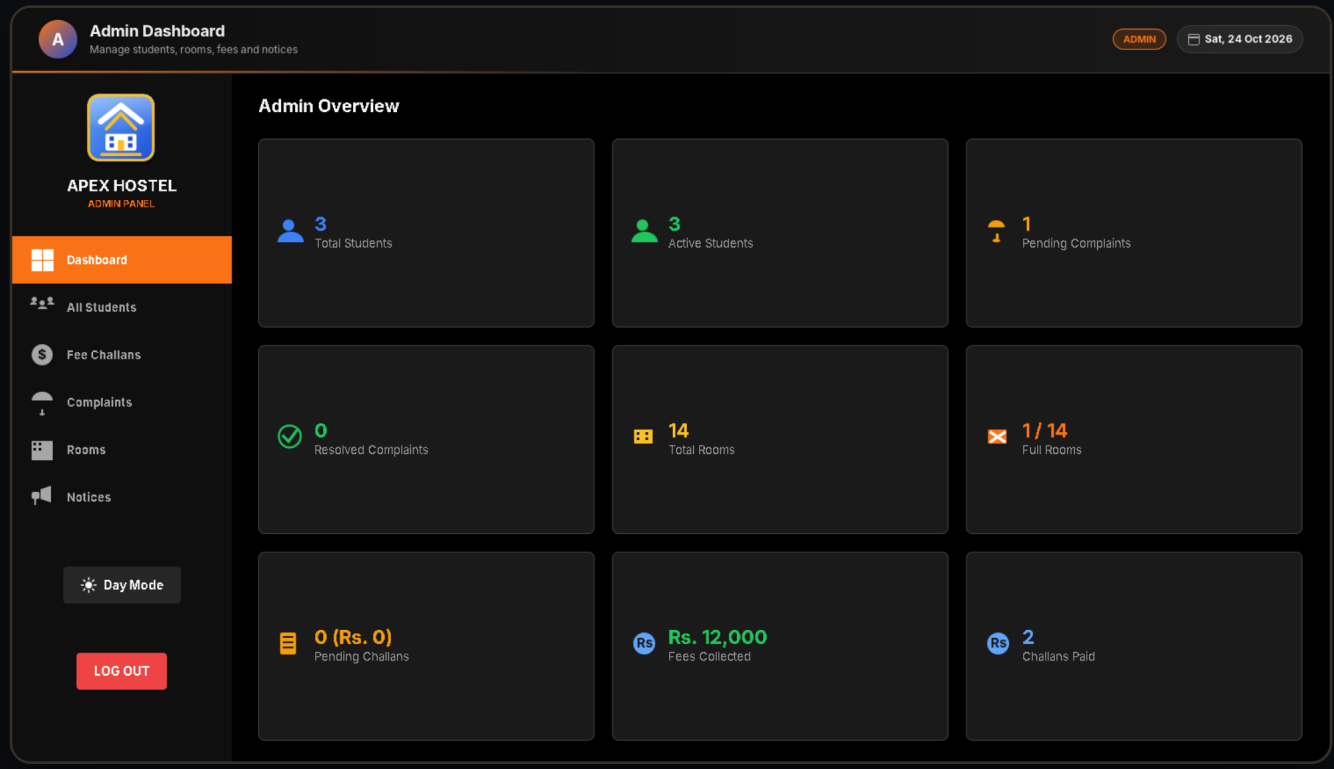
- Every room with type, rent and seats
- Available and Full shown by colour and word
- Your own room is highlighted
- Filter: only rooms with free seats
- Students cannot allocate. The admin does.



ADMIN SIDE

Admin dashboard

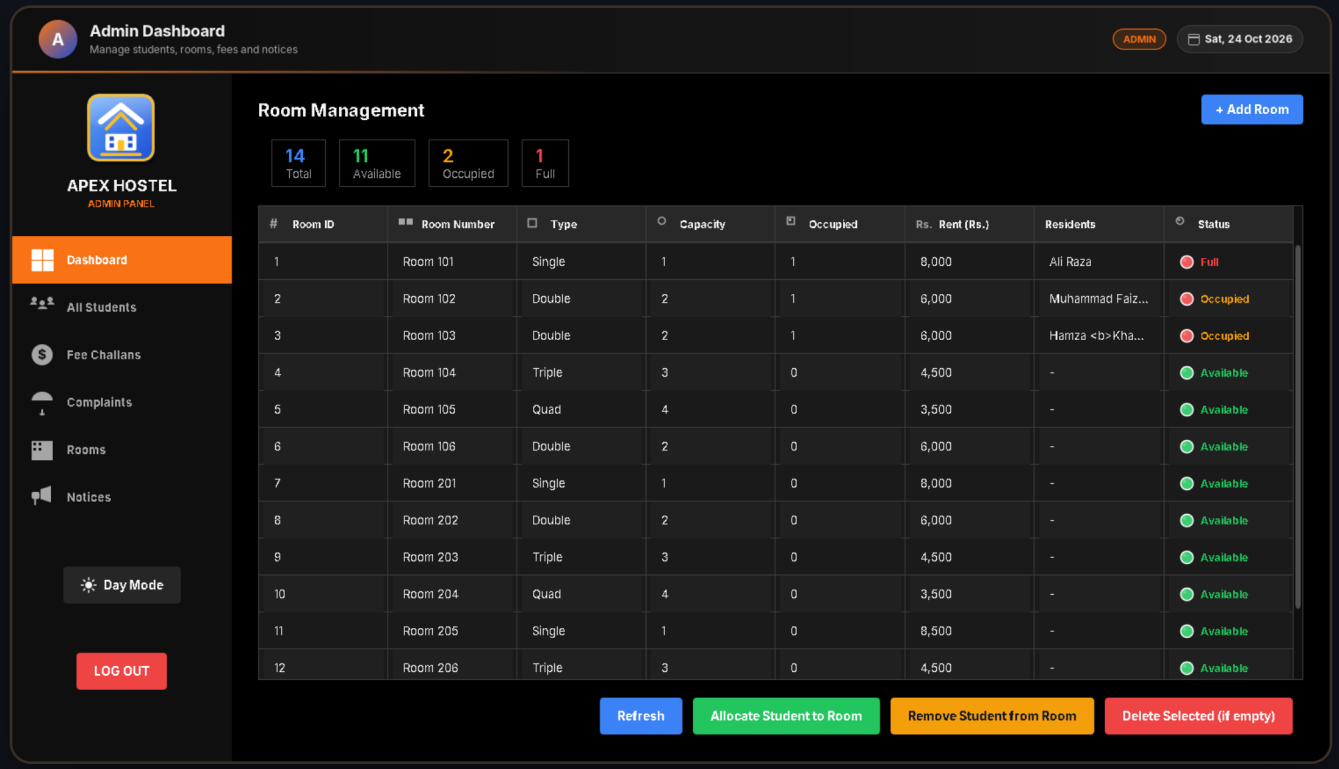
- Live counts of students, rooms and complaints
- Fees collected and pending challans
- Pages for students, fees, complaints, rooms and notices
- Orange accent marks the admin panel



ADMIN SIDE

Only the admin allocates rooms

- Allocate or remove a student in one click
- Full rooms are refused automatically
- Residents column shows who lives where
- Status updates the moment a room changes

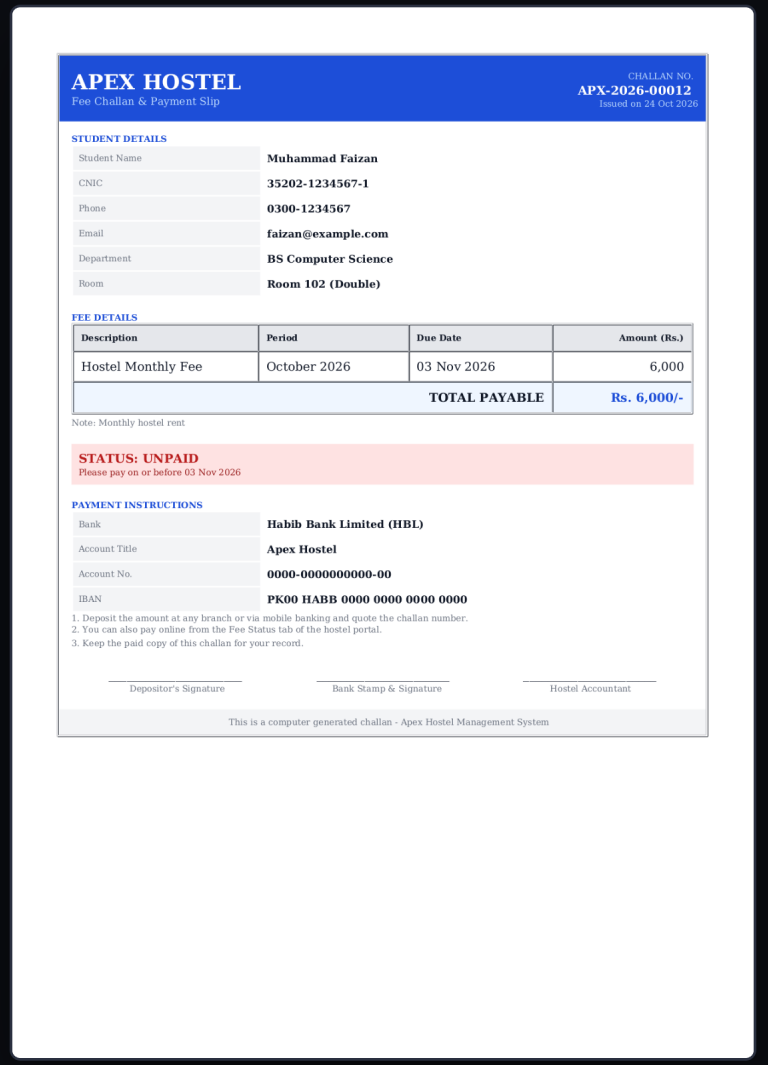


# Room ID	Room Number	Type	Capacity	Occupied	Rs. Rent (Rs.)	Residents	Status
1	Room 101	Single	1	1	8,000	Ali Raza	Full
2	Room 102	Double	2	1	6,000	Muhammad Faiz...	Occupied
3	Room 103	Double	2	1	6,000	Hamza <b>Kha...	Occupied
4	Room 104	Triple	3	0	4,500	-	Available
5	Room 105	Quad	4	0	3,500	-	Available
6	Room 106	Double	2	0	6,000	-	Available
7	Room 201	Single	1	0	8,000	-	Available
8	Room 202	Double	2	0	6,000	-	Available
9	Room 203	Triple	3	0	4,500	-	Available
10	Room 204	Quad	4	0	3,500	-	Available
11	Room 205	Single	1	0	8,500	-	Available
12	Room 206	Triple	3	0	4,500	-	Available

FEES

One-page fee challan

- Admin creates one challan, or all students for a month
- Shows challan number, student, room, amount and due date
- Bank payment instructions and signature boxes
- Prints on exactly one A4 page
- Download PDF button saves the same single page



APEX HOSTEL
Fee Challan & Payment Slip

CHALLAN NO.
APX-2026-00012
Issued on 24 Oct 2026

STUDENT DETAILS

Student Name	**Muhammad Faizan**
CNIC	**35202-1234567-1**
Phone	**0300-1234567**
Email	**faizan@example.com**
Department	**BS Computer Science**
Room	**Room 102 (Double)**

FEE DETAILS

Description	Period	Due Date	Amount (Rs.)
Hostel Monthly Fee	October 2026	03 Nov 2026	6,000
		TOTAL PAYABLE	**Rs. 6,000/-**

Note: Monthly hostel rent

STATUS: UNPAID
Please pay on or before 03 Nov 2026

PAYMENT INSTRUCTIONS

Bank	**Habib Bank Limited (HBL)**
Account Title	**Apex Hostel**
Account No.	**0000-0000000000-00**
IBAN	**PK00 HABB 0000 0000 0000 0000**

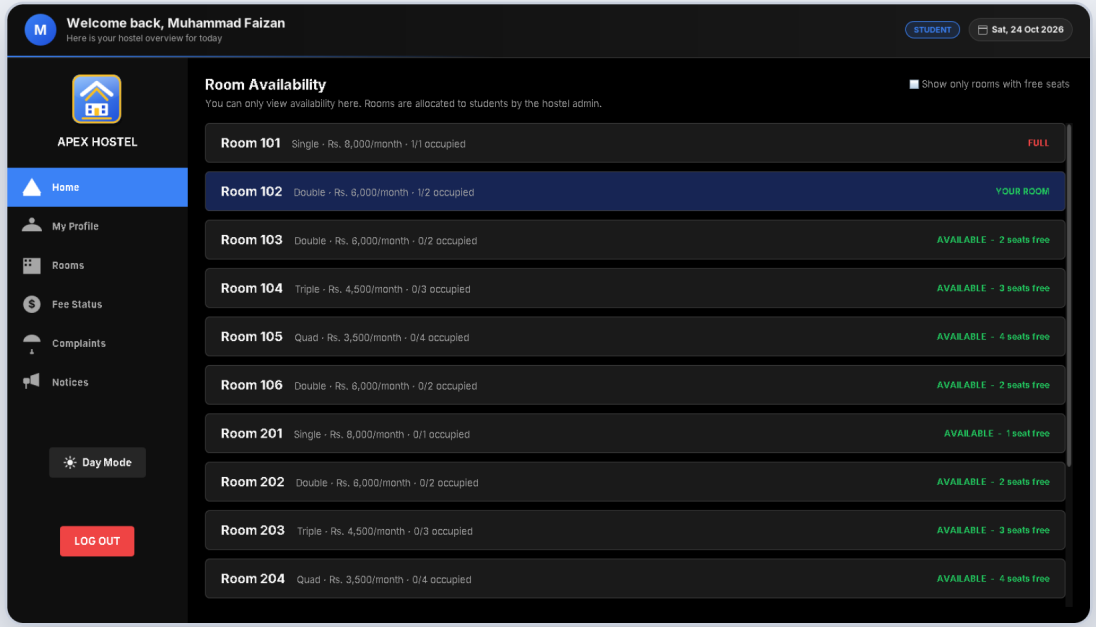
1. Deposit the amount at any branch or via mobile banking and quote the challan number.
2. You can also pay online from the Fee Status tab of the hostel portal.
3. Keep the paid copy of this challan for your record.

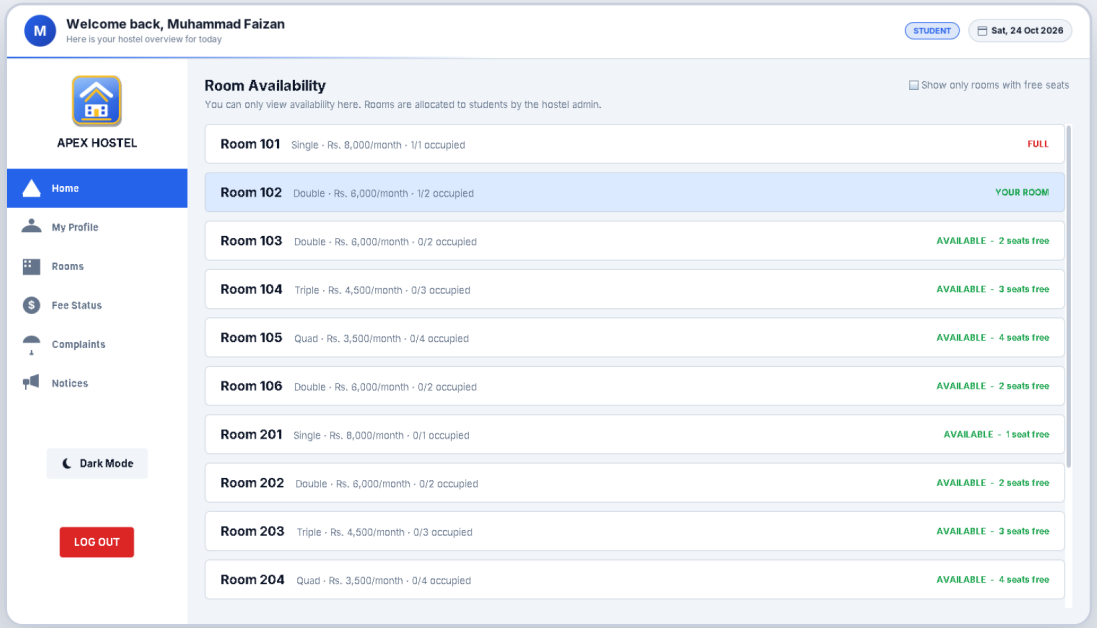
Depositor's Signature | Bank Stamp & Signature | Hostel Accountant

This is a computer generated challan - Apex Hostel Management System

LOOK AND FEEL

Dark mode and Day mode





Switch from the sidebar or login page. The choice is remembered. Thank you, any questions?